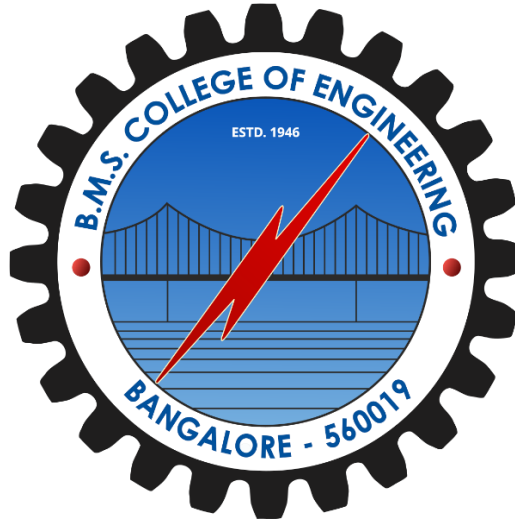


BMS College of Engineering



(AUTONOMOUS COLLEGE AFFILIATED TO VTU BELGAM) BULL TEMPLE ROAD ,
BASAVANGUDI BANGLORE-560019

PROJECT BASED LEARNING (AAT) ACTIVE FIN STABILIZATION SYSTEM OF ROCKET MISSILE'S

COUSE TITLE : Measurements and Instrumentation

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1. Abstract

Rocket flight stability is a critical requirement for achieving safe, accurate, and reliable trajectories. This project presents the design and implementation of an **active fin stabilization system** based on real-time feedback control. An **MPU-6050 motion sensor**, integrating a three-axis accelerometer and gyroscope, is used to continuously measure the rocket's orientation. Sensor data are fused using a **complementary filter** to obtain a stable and accurate attitude estimate. A **PID (Proportional–Integral–Derivative) control algorithm** processes the orientation error and generates corrective signals to actuate servo-driven fins. These active fins dynamically respond to external disturbances such as wind gusts and thrust misalignment, ensuring improved directional stability during flight. The proposed system demonstrates significantly enhanced stability and responsiveness when compared to conventional passive fin designs, highlighting the effectiveness of active control techniques in rocket stabilization applications.

2. Introduction

Rocket stability is a critical factor in ensuring safe, accurate, and efficient flight performance. During launch and ascent, rockets are subjected to various disturbances such as wind gusts, thrust misalignment, and aerodynamic forces, which can cause unwanted rotation or deviation from the intended trajectory. Traditional rockets rely on passive fins for stabilization; however, passive systems cannot actively respond to sudden disturbances or changing flight conditions. To overcome these limitations, active fin stabilization systems are increasingly used in modern aerospace applications.

This project focuses on the design and implementation of an active fin stabilization system for a rocket using sensor-based feedback control. The system continuously monitors the rocket's orientation and dynamically adjusts the fins to maintain a stable flight path. An MPU-6050 sensor, which combines a 3-axis accelerometer and a 3-axis gyroscope, is used to measure angular motion and acceleration in real time. These measurements allow the system to estimate the rocket's attitude accurately during flight.

A PID (Proportional–Integral–Derivative) control algorithm is employed to compute corrective actions based on the difference between the desired and actual orientation of the rocket. The controller ensures fast response, minimal oscillation, and stable control of the fins. Servo motors driven through a dedicated servo controller actuate the fins according to the control output. The control logic and data processing are implemented using Python, with PyCharm as the development environment, enabling efficient testing and debugging.

Overall, this project demonstrates how active control systems can significantly improve rocket stability compared to passive methods, making it a valuable study in control systems, embedded electronics, and aerospace engineering fundamentals.

3. Objectives of the Project

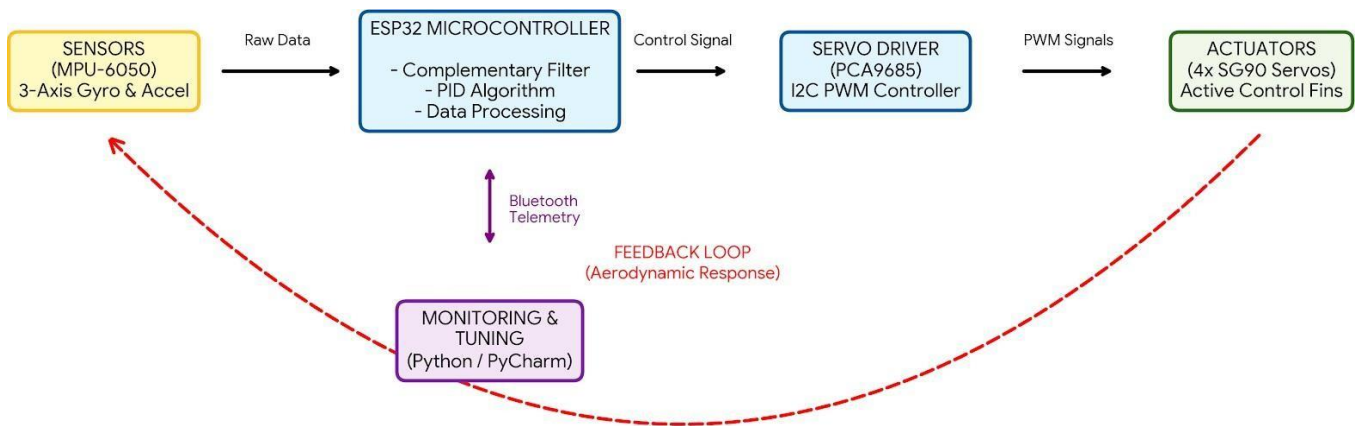
The main objective of this project is to design and implement an active fin stabilization system for a rocket to improve flight stability and control. The project aims to measure the rocket's real-time orientation using the MPU-6050 motion sensor and accurately estimate attitude through sensor data fusion. Another objective is to develop and implement a PID control algorithm that computes corrective actions based on orientation errors. The system also aims to control servo-driven fins effectively to counteract external disturbances such as wind gusts and thrust misalignment. Overall, the project seeks to demonstrate improved stability and performance compared to conventional passive fin systems.

4. Problem Statement

Maintaining stable and accurate flight trajectories in rockets is a significant challenge due to external disturbances such as wind gusts, thrust misalignment, and aerodynamic uncertainties. Conventional passive fin stabilization systems provide limited control and cannot actively respond to these dynamic disturbances during flight. As a result, rockets may experience unwanted rotation, oscillations, or deviation from the intended path, reducing accuracy and safety. Therefore, there is a need for an active stabilization system that can continuously monitor the rocket's orientation and dynamically adjust control surfaces in real time to correct disturbances and ensure stable flight performance.

5. Working Principle

The active fin stabilization system works by continuously sensing the rocket's orientation using the MPU-6050 sensor. Sensor data from the accelerometer and gyroscope are combined to obtain a stable angle estimate. This angle is compared with the desired reference angle, and the error is processed by a PID controller. The controller output drives servo motors, which adjust the fins in real time to counteract disturbances and maintain stable flight.



Block Diagram Breakdown

- **Sensors (MPU-6050):** Acts as the rocket's "inner ear," detecting angular changes and movement.
- **ESP32 Microcontroller:** The main processing unit that implements the **Complementary Filter** and **PID algorithm** to calculate real-time corrections.
- **Python/PyCharm & Bluetooth:** This block represents your development and monitoring station. Via **Bluetooth**, you can receive live telemetry and wirelessly tune the PID constants during testing.
- **PCA9685 & Servos:** The hardware interface that translates digital signals into precise physical movement of the four **SG90 servos**.
- **Active Fins & Feedback Loop:** The movement of the fins changes the aerodynamics, which the sensors immediately detect, closing the loop and maintaining stability.

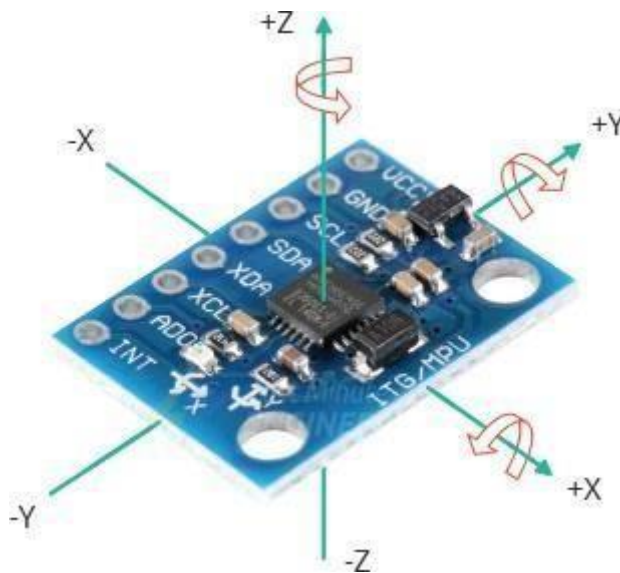
HARDWARE AND SOFTWARE DESCRIPTIONS

5.1 Sensor System – MPU-6050(HARDWARE DESCRIPTIONS)

The MPU-6050 is a widely used MEMS-based motion sensor that integrates a 3-axis accelerometer and a 3-axis gyroscope on a single chip. Its compact size, low power consumption, and high accuracy make it suitable for real-time motion and orientation sensing in embedded control systems such as rocket stabilization.

The accelerometer measures linear acceleration along the X, Y, and Z axes using capacitive sensing elements that detect changes caused by movement or tilt with respect to gravity. The gyroscope measures angular velocity along the same three axes using vibrating MEMS structures based on the Coriolis effect, allowing detection of rotational motion.

The sensor provides six raw parameters: AcX , AcY , and AcZ represent acceleration along the three axes, while GyX , GyY , and GyZ represent angular velocity about those axes. These parameters are essential for determining the rocket's orientation, rotation, and motion. By continuously monitoring these values, the MPU-6050 enables accurate attitude sensing, which is critical for effective active fin control and overall flight stability.



5.1.1 Sensor Construction (MPU-6050)

The MPU-6050 is a compact MEMS-based motion sensor that integrates a **3-axis accelerometer** and a **3-axis gyroscope** on a single chip. The accelerometer is constructed using microscopic capacitive structures that detect linear acceleration along the X, Y, and Z axes by sensing changes in capacitance caused by motion. The gyroscope consists of vibrating MEMS elements that measure angular velocity based on the Coriolis effect when the sensor rotates.

Both sensors are embedded on a silicon die along with signal conditioning circuits and an internal Digital Motion Processor (DMP). The device also includes an I²C communication interface for data transfer and an internal temperature sensor for calibration. This compact construction allows accurate, high-speed motion sensing while consuming low power, making the MPU-6050 suitable for real-time attitude measurement in active fin stabilization systems.

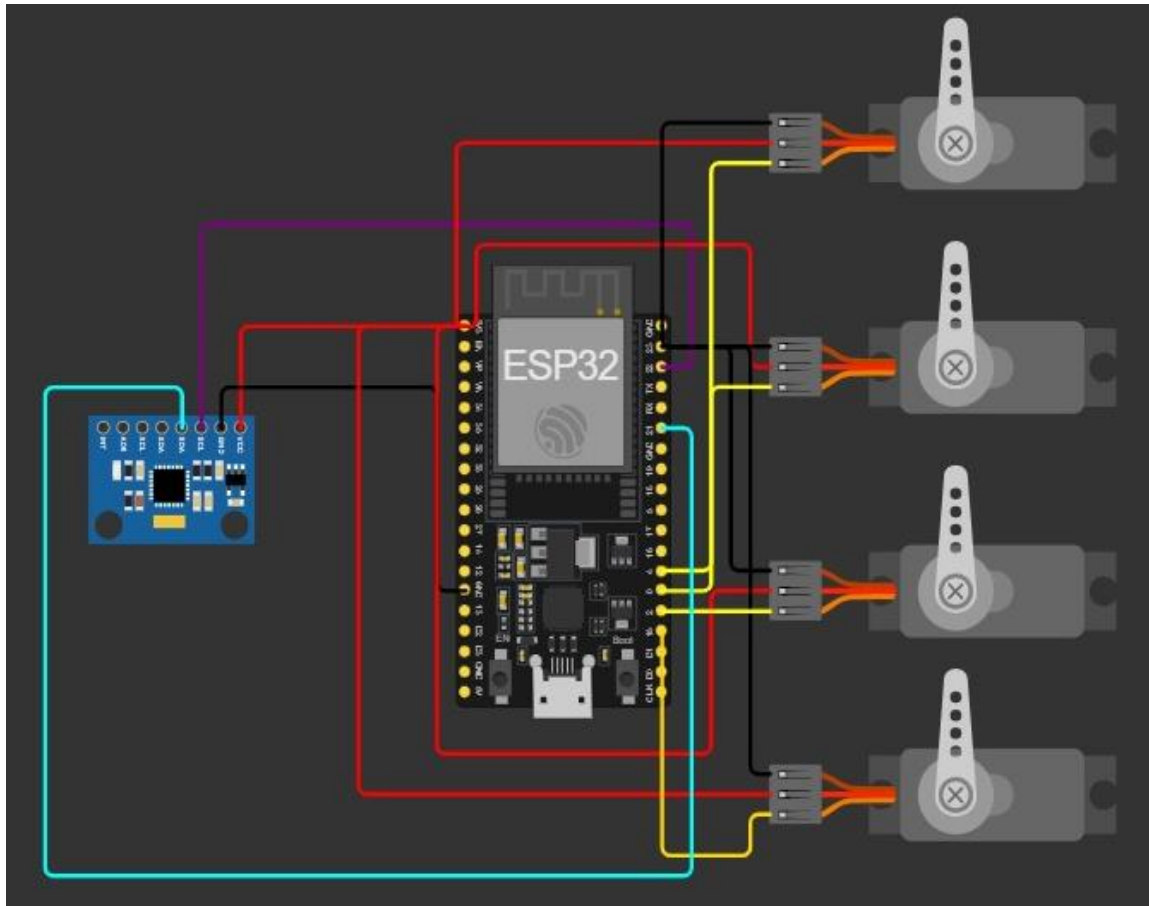
5.2 Software Description

The software for the active fin stabilization system is responsible for sensor data acquisition, attitude estimation, control computation, and actuator control. The entire control logic is implemented using **Python**, selected for its simplicity, readability, and strong support for mathematical computations and sensor data processing. The software continuously reads raw acceleration and angular velocity data from the **MPU-6050 motion sensor**.

To obtain a reliable estimate of the rocket's orientation, a **complementary filter** is implemented in software to fuse accelerometer and gyroscope data, reducing noise and long-term drift. Based on the estimated attitude, a **PID (Proportional–Integral–Derivative) control algorithm** calculates the required correction to minimize orientation error. The computed control signals are then translated into appropriate commands for the servo motors through the servo driver.

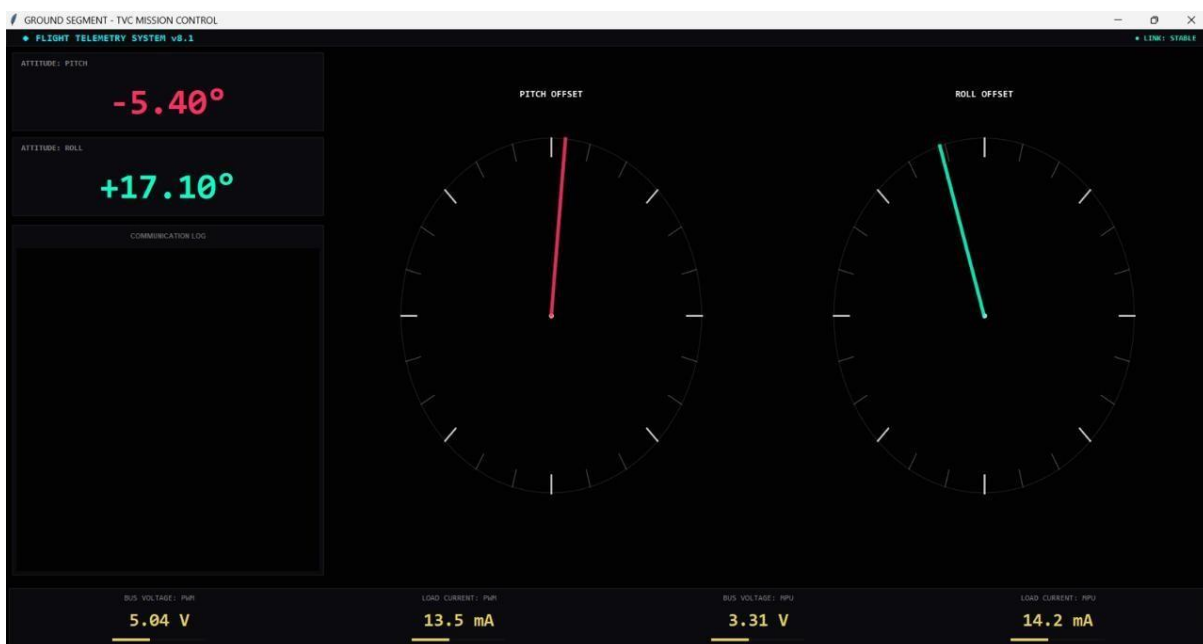
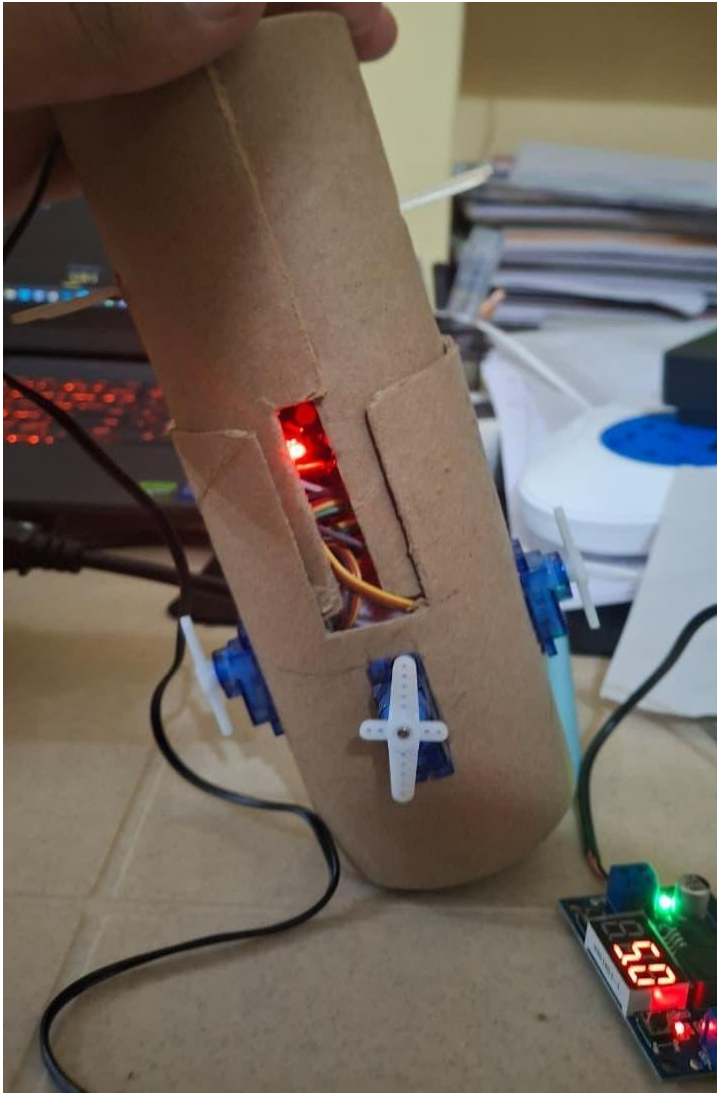
All software development, testing, and debugging were carried out using the **PyCharm Integrated Development Environment (IDE)**, which provided tools such as code execution, error detection, and debugging support, enabling efficient development of the stabilization system.

5.3 CIRCUIT CONNETIONS



5.4 PROJECT PICS







FINAL FRONT VIEW OF THE PROJECT

6. Angle Estimation & Complementary Filter

Accurate angle estimation is essential for maintaining rocket stability. The gyroscope in the MPU-6050 provides fast and smooth angular measurements, but its output gradually drifts due to accumulated integration errors. The accelerometer, while offering an absolute tilt reference based on gravity, is sensitive to noise, vibration, and acceleration during flight. To overcome these limitations, a complementary filter is used to combine both sensor outputs. This filter relies on the gyroscope for short-term motion accuracy and uses accelerometer data to correct long-term drift

- **Gyroscope Output**
 - Provides smooth and fast angle changes
 - Suffers from drift over time due to integration errors
- **Accelerometer Output**
 - Gives absolute tilt reference using gravity
 - Affected by noise, vibration, and acceleration
- **Complementary Filter**
 - Combines gyroscope and accelerometer data
 - Gyro handles short-term motion
 - Accelerometer corrects long-term drift
- **Mathematical Relation**

The formula for the filtered angle is expressed as:

$$Angle = \alpha \cdot (Angle_{gyro}) + (1 - \alpha) \cdot (Angle_{accel})$$

- **Result / Advantage**
 - Smooth, stable, and accurate angle estimation
 - Reliable attitude sensing for PID control

6. Control System – PID Controller

The PID (Proportional–Integral–Derivative) controller is the core control mechanism used in the active fin stabilization system. It is a feedback-based control technique that continuously adjusts the system output to maintain the desired rocket orientation. PID control is widely used due to its simplicity, reliability, and effectiveness in dynamic systems.

The controller operates by calculating the error, which is the difference between the desired reference angle (usually 0°, representing straight flight) and the actual angle measured by the sensors. This error indicates how much the rocket has deviated from its intended path.

Based on this error, the PID controller generates a corrective control signal that determines the movement of the fins. The proportional term reacts to the present error, the integral term accounts for accumulated past errors, and the derivative term predicts future error trends.

The PID control equation can be written neatly as:

$$\text{PID Output} = K_p e(t) + K_i \int e(t) dt + K_d \frac{de(t)}{dt}$$

where:

- $e(t)$ is the error signal
- K_p is the proportional gain
- K_i is the integral gain
- K_d is the derivative gain

7. Results and Discussion

The active fin stabilization system demonstrated a clear improvement in maintaining rocket stability when subjected to external disturbances. During testing, the system responded effectively to simulated disturbances such as sudden angular deviations, representing wind gusts and thrust misalignment. The MPU-6050 sensor provided continuous orientation feedback, allowing the control system to detect deviations promptly.

The PID controller generated appropriate corrective signals, enabling the servo-driven fins to counteract disturbances in real time. As a result, the rocket quickly returned to its desired orientation with minimal overshoot and oscillation. Compared to passive fins, which rely solely on aerodynamic forces and cannot actively correct disturbances, the active fin system showed significantly faster response and improved stability.

Additionally, fin movements were observed to be smooth and well-controlled, indicating proper tuning of the PID parameters. The combination of sensor fusion and closed-loop control reduced erratic behavior and ensured consistent performance. Overall, the results confirm that the active fin stabilization system provides superior control and stability compared to conventional passive fin designs.

8. Advantages

- Provides **real-time correction of disturbances** such as wind gusts and thrust misalignment during flight.
- Enhances **flight accuracy** by maintaining the rocket's desired orientation and trajectory.
- Offers **better stability compared to passive fin systems**, which cannot actively respond to dynamic conditions.
- Enables **smooth and controlled fin movements** through closed-loop PID control.
- The system is **scalable and adaptable** for advanced rockets and more complex guidance and navigation applications.

9. Future Scope

The active fin stabilization system can be further enhanced by incorporating advanced sensor fusion techniques such as the Kalman filter to improve attitude estimation accuracy. Integration of GPS and altitude sensors can enable full navigation and trajectory control. High-speed and high-torque actuators may be used for better performance in high-velocity rockets. The system can also be extended to autonomous flight control with adaptive or self-tuning PID algorithms. Additionally, implementing wireless telemetry and data logging would allow real-time monitoring and post-flight analysis, making the system suitable for advanced aerospace research and experimental rocket applications.

10. Present Trends and Industrial Adoption

Active stabilization and guidance systems are currently being developed and utilized by leading aerospace and defense organizations such as **NASA, Boeing, and the Defence Research and Development Organisation (DRDO)**. These organizations employ advanced active control techniques, including fin-based stabilization, thrust vector control, and real-time feedback systems, to ensure precise trajectory control and mission reliability. Such technologies are widely used in launch vehicles, guided missiles, and reusable space systems. The increasing adoption of active stabilization systems highlights their importance in modern aerospace engineering and reflects the growing demand for accurate, safe, and autonomous flight control solutions.

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Game changer: India test-fires Pralay missiles in rapid succession off Odisha coast

News Desk

31 December 2025, 04:56 PM IST



Pralay missile

New Delhi: India on Wednesday successfully test-fired two Pralay missiles in quick succession off the coast of Odisha, marking another milestone for the country's indigenous missile development program.

The above image is news report of DRDO PRALAY missiles which is guided missiles.

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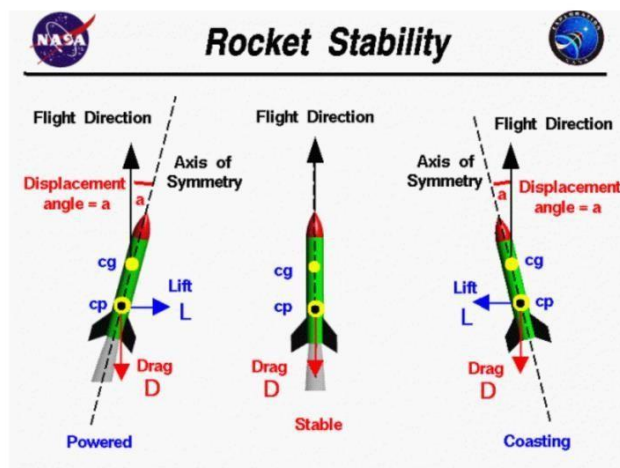
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[Why is Flight Direction Vertical?](#)

[De-Stabilizing Force](#)

[How to Determine the Stability?](#)

Rocket Stability



The above image is of NASA research page where they are using this technique for rocket stability

Global Avionics Round-Up from Aircraft Value News (AVN)

A350-1000 Takes the Lead in Autonomous Flying

By [John Persinos](#) | October 3, 2024

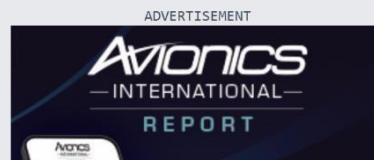
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Airbus A350-1000, Aircraft Value News (AVN), autonomous



Global Avionics Round-Up from Aircraft Value News (AVN)

The increasing prevalence of autonomous aviation aligns with broader technological advancements within avionics. The A350-1000 is a test case.



NEWS report from avionics international stating that airbus is using billions of dollars in automated systems

11. Conclusion

This project successfully demonstrated the design and implementation of an active fin stabilization system for rocket flight control. By utilizing the MPU-6050 motion sensor, accurate real-time orientation data was obtained through sensor fusion techniques. The implementation of a PID control algorithm enabled precise and smooth control of servo-driven fins, effectively counteracting external disturbances such as wind gusts and thrust misalignment. Compared to conventional passive fin systems, the proposed active stabilization approach showed significantly improved stability, faster response, and better flight accuracy. The project highlights the importance of feedback control systems in aerospace applications and provides a strong foundation for further development of advanced rocket guidance and stabilization technologies.

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